

1. TENTANG AUTO-EDA DASHBOARD

Auto-EDA Dashboard (DS Generator) adalah platform analisis data otomatis berbasis web yang dikembangkan oleh **Kelompok 2 Data Science ITSB**. Aplikasi ini dirancang untuk membantu analis data, manajer, dan pengambil keputusan bisnis dalam melakukan: (1) unggah dan validasi dataset, (2) audit kesehatan data (quality audit), (3) pembersihan data interaktif (data cleaning), (4) perhitungan statistik deskriptif dan lanjutan, (5) visualisasi pola data secara interaktif, serta (6) perumusan insight dan rekomendasi strategis berbasis data secara otomatis dan real-time.

2. ANGGOTA TIM PENGEMBANG (KELOMPOK 2 ITSB)

Nama Lengkap	NIM	Peran / Fokus Analisis
Carol Dupino Pereira	52250051	Descriptive & Advanced Statistics Engine
Refantanur Husnul Haqib	52250052	Visualizations & Dynamic Plotly Dashboard
Cahaya Medina Semidang	52250053	Data Preprocessing & Sanitizer Module
Raihania Syah Putri	52250054	Time Series Forecasting & Trends Panel
Cloise Shafira	52250044	Smart Insights Generation Algorithm
Adinda Adelia Futri	52250055	Reporting System PDF/Excel & Security Sanitization

3. DESKRIPSI DATA & RINGKASAN EKSEKUTIF

Nama File Dataset:	supermarket_sales.csv
Waktu Analisis:	14 June 2026, 13:34:35
Status Kebersihan:	RAW DATA (Perlu Pembersihan)
Total Baris (Observasi):	1000
Total Kolom (Variabel):	17
Variabel Numerik:	7
Variabel Kategorikal:	7

Missing Cells	Duplicate Rows	IQR Outliers
0 (0.0%)	0	36

Identifikasi Masalah Kualitas Data:

- 36 outlier terdeteksi (metode IQR)
- 5 kolom memiliki inkonsistensi format teks
- 6 kolom kemungkinan tidak relevan (ID/free text)

Detail Kesehatan Per Kolom:

Kolom	Tipe	Missing (%)	Unique	Outliers	Status
Invoice ID	object	0 (0.0%)	1000	0	Possible ID/free text
Branch	object	0 (0.0%)	3	0	OK
City	object	0 (0.0%)	3	0	OK
Customer type	object	0 (0.0%)	2	0	OK
Gender	object	0 (0.0%)	2	0	OK
Product line	object	0 (0.0%)	6	0	OK
Unit price	float64	0 (0.0%)	943	0	OK
Quantity	int64	0 (0.0%)	10	0	OK
Tax	float64	0 (0.0%)	990	9	9 outliers, Possible ID/free text
Total	float64	0 (0.0%)	990	9	9 outliers, Possible ID/free text
Date	object	0 (0.0%)	89	0	OK
Time	object	0 (0.0%)	506	0	OK
Payment	object	0 (0.0%)	3	0	OK
cogs	float64	0 (0.0%)	990	9	9 outliers, Possible ID/free text
gross margin percentage	float64	0 (0.0%)	1	0	Constant column
gross income	float64	0 (0.0%)	990	9	9 outliers, Possible ID/free text
Rating	float64	0 (0.0%)	61	0	OK

3a. ALUR PROSES DATA CLEANING (PIPELINE)

Dataset diproses melalui 5 tahapan alur data cleaning di backend engine:

- 1. **Audit Kesehatan (Data Profiling):** Mendeteksi sel kosong (NaN), baris duplikat, inkonsistensi teks, dan outlier numerik (IQR).
- 2. **Standarisasi Teks:** Menghilangkan spasi berlebih dan menyeragamkan bentuk huruf (Title/Lower/Upper Case).
- 3. **Imputasi Data Hilang:** Mengisi sel kosong dengan mean/median/modus atau menghapus baris jika kerusakan ekstrem.
- 4. **Pembersihan Outlier (IQR Capping):** Menetralkan nilai ekstrim menggunakan batas IQR.
- 5. **Drop Kolom Tidak Relevan:** Menghapus kolom variansi nol atau rasio keunikan terlalu tinggi (ID/teks acak).

Metrik	Sebelum (Raw)	Sesudah (Cleaned)	Perubahan
Baris Data	1000	1000	Dihapus: 0 baris
Kolom Data	17	17	Dihapus: 0 kolom
Sel Kosong	0 (0.0%)	0 (0.0%)	Dibersihkan: 0 sel

Log Tindakan Cleaning:

• Tidak ada missing value ditemukan pada kolom yang dipilih.

3b. Statistik Deskriptif — Variabel Numerik

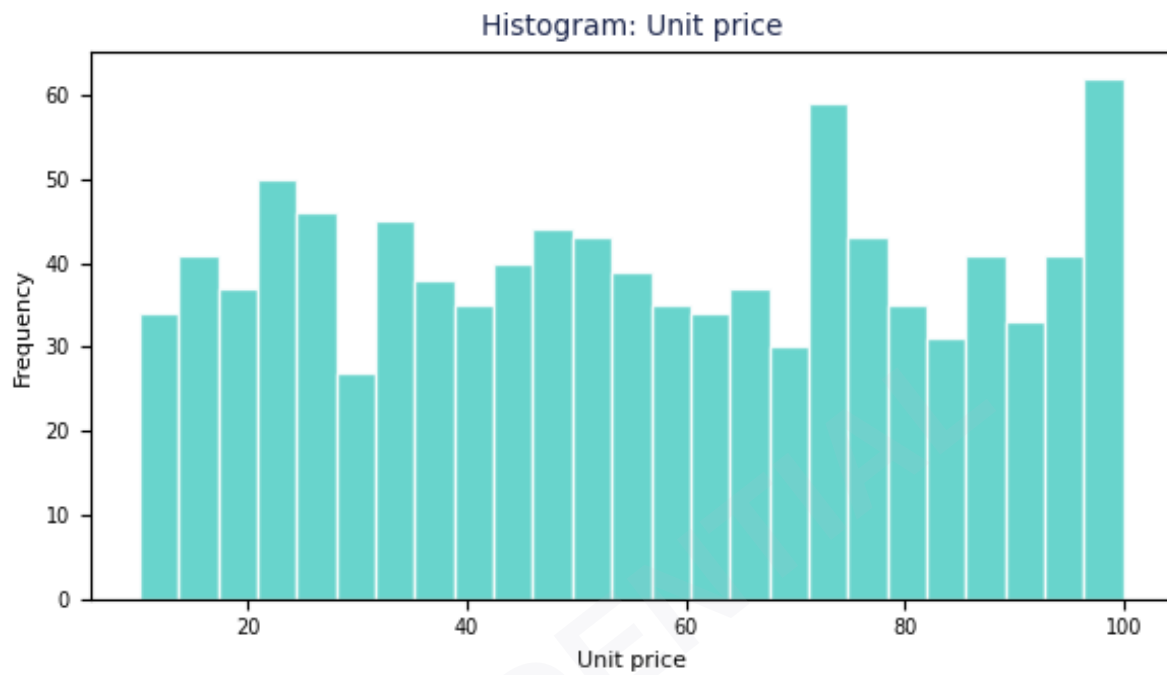
Kolom	Mean	Median	Min	Max	Std Dev	Skewness	Normality
Unit price	55.67	55.23	10.08	99.96	26.49	0.0071	Not Normal
Quantity	5.51	5.0	1.0	10.0	2.92	0.0129	Not Normal
Tax	15.38	12.09	0.51	49.65	11.71	0.8926	Not Normal
Total	322.97	253.85	10.68	1042.65	245.89	0.8926	Not Normal
cogs	307.59	241.76	10.17	993.0	234.18	0.8926	Not Normal
gross income	15.38	12.09	0.51	49.65	11.71	0.8926	Not Normal
Rating	6.97	7.0	4.0	10.0	1.72	0.009	Not Normal

3c. Statistik Deskriptif — Variabel Kategorikal

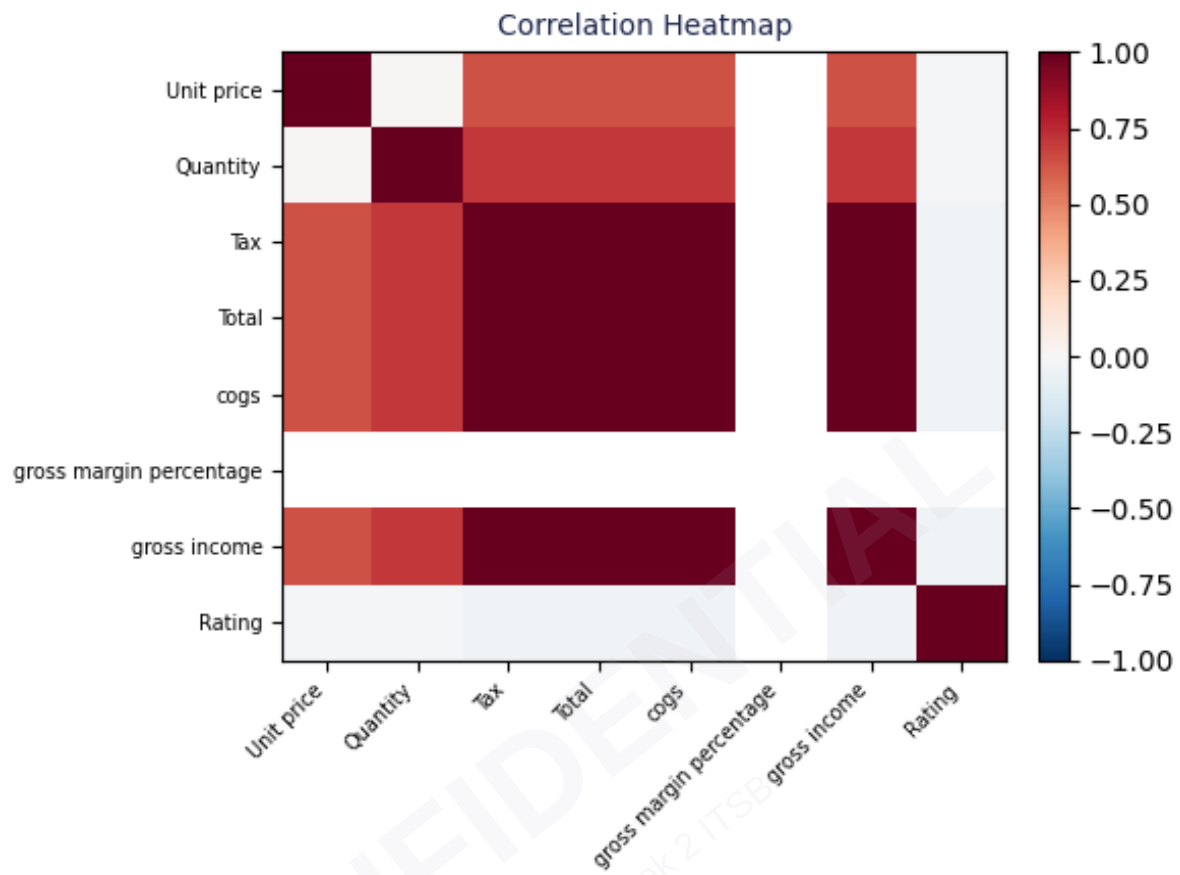
Kolom	Unique	Mode	Mode Freq	Mode %	Missing	Missing %
Branch	3	A	340	34.0%	0	0.0%
City	3	Yangon	340	34.0%	0	0.0%
Customer type	2	Member	501	50.1%	0	0.0%
Gender	2	Female	501	50.1%	0	0.0%
Product line	6	Fashion accessories	178	17.8%	0	0.0%
Payment	3	Ewallet	345	34.5%	0	0.0%
gross margin percentage	1	4.761904762	1000	100.0%	0	0.0%

4. VISUALISASI KUNCI (HISTOGRAM & HEATMAP)

4a. Histogram (Distribusi Variabel Numerik)



4b. Heatmap Korelasi (Hubungan Antar Variabel)



5. TEMUAN KUNCI & INSIGHT OTOMATIS

Excellent Data Quality — Zero Missing Values

No missing values found at all. The dataset is clean and ready for statistical analysis or machine learning without additional preprocessing.

Dataset: 1,000 Rows × 17 Columns

Dataset size is sufficiently representative for reliable statistical analysis. Consists of 7 numeric and 7 categorical columns.

Highest Average Value: Total

Column **Total** has the highest average value of **322.97**. Median: 253.85, Std: 245.89.

Most Outliers: Tax

Column **Tax** has the most outliers: **9 data points (0.9%)** based on IQR method ($\pm 1.5 \times \text{IQR}$). Check whether these outliers are data errors or valid extreme values.

Highest Std Deviation: Total

Column **Total** has the largest standard deviation of **245.8853** (CV=76.1%). Very high variability on this column indicates a wide data spread.

Strongest Correlation: Tax ↔ gross income (r=1.0)

Very Strong (positive) correlation found between **Tax** and **gross income** with $r = 1.0$. This relationship is highly significant and worth investigating further. ($R^2 = 1.0$)

Normality Test — 0/7 Columns Normal

NOT normal columns: Unit price, Quantity, Tax (+4 more). Non-normal columns (100.0%) require non-parametric tests (Mann-Whitney, Kruskal-Wallis) or data transformation (log, sqrt, Box-Cox) before parametric analysis.

Categorical Balance (Branch): Balanced

Category distribution is fairly even. Most frequent category: "A" (34.0%). Total 3 unique categories in column Branch.

Trend: Menurun (Downward) ↓

Variabel Unit price menunjukkan tren menurun (downward) ↓ dengan slope=-0.0108 per periode (Hourly). $R^2=0.001$ — tren belum signifikan ($p \geq 0.05$).

Volatilitas: Sedang (CV 10–30%)

Koefisien variasi (CV) = 13.9%. Nilai ini menunjukkan tingkat fluktuasi yang sedang (cv 10–30%) pada Unit price sepanjang periode waktu.

Peak & Trough — Unit price

Nilai tertinggi: 81.40 pada 2019-02-11. Nilai terendah: 37.86 pada 2019-02-13. Range: 43.54.

Tidak Ada Pola Musiman Jelas

Tidak ditemukan autocorrelation signifikan pada lag standar (7, 12, 30). Data Unit price tampak tidak memiliki pola musiman yang kuat.

Further Analysis Recommendations

Based on the structure of this dataset, suggested techniques: One-Way ANOVA / T-Test to compare means across categorical groups; Linear / Logistic Regression for prediction and modeling; Chi-Square + Cramér's V for categorical variable association; Non-Parametric Tests (Mann-Whitney, Spearman) for non-normal columns; Principal Component Analysis (PCA) for dimensionality reduction.

6. REKOMENDASI STRATEGIS PENGAMBILAN KEPUTUSAN

R1. Investigasi Nilai Anomali (Outliers): Ditemukan data pencilan ekstrem. Disarankan tim operasional memvalidasi apakah angka tersebut valid (fluktuasi bisnis riil) atau human error. Lakukan capping/truncation sebelum pemodelan prediktif lanjutan.

R2. Pembersihan Berkala (Data Governance): Jadwalkan proses pembersihan data secara rutin menggunakan Auto-EDA Dashboard sebelum laporan akhir bulan diekspor ke departemen eksekutif, guna menjamin keputusan dibuat berdasarkan data higienis.